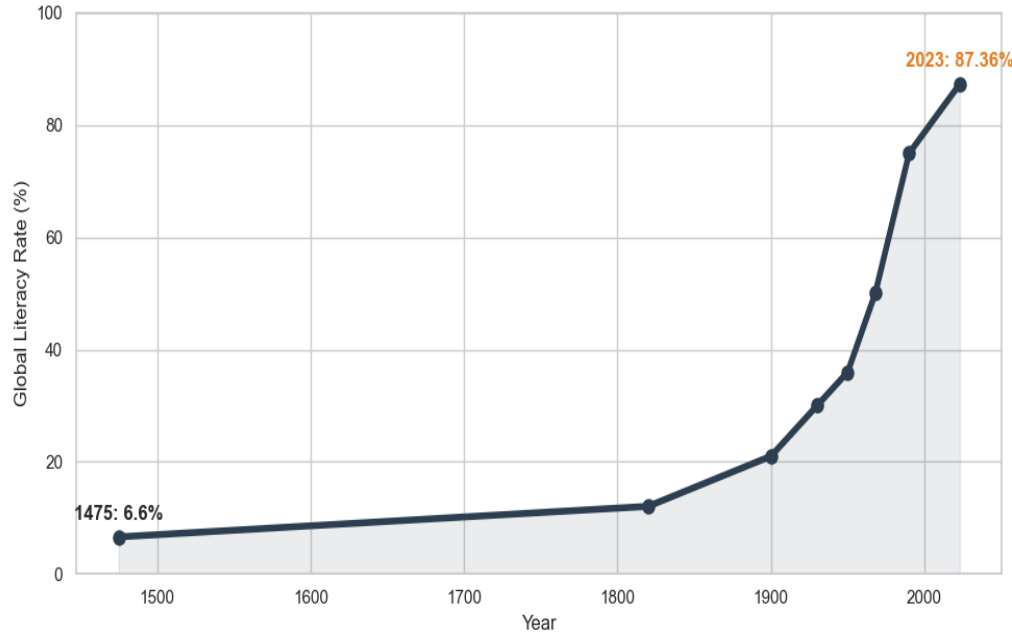


Data Visualization Pitch

Student Individual Assignment

THE GREAT LITERACY INVERSION
550 Years of Global Progress (1475-2023)



This chart is created with
Python(Matplotlib/Seaborn)

Literacy Rates Around the World: Share of adults who can read and write a simple statement about their everyday life

Student name: Amro Emad Hassan Askar

Student ID: 946807

Student e-mail address: a.askar@campus.unimib.it

Research questions

How has the global landscape of literacy transformed over the last 500 years, and what do the specific "tipping points" and "statistical anomalies" reveal about the uneven pace of educational development across different regions?

- The Weighting Paradox: Why does the global "tipping point" (50% literacy) shift by five decades when moving from unweighted country averages (1910s) to population-weighted global estimates (1960s)?
- Which countries or regions are still falling short of the global average in the 21st century?

About Data

Dataset "Literate and illiterate world population [Link](#)": Dataset from Our World in Data. This file contains historical literacy rates beginning in 1475 for specific countries and global estimates from 1820 to 2023.

License: The dataset is shared under Creative Commons BY (CC-BY) terms. This allows me to make free use, transformation, and sharing of this data for this academic assignment.

Quality and completeness concerns:

- ❖ **Data Sparsity:** The date range of 1475-2023 represents a very sparse data series in the early centuries, but the data becomes globally consistent only after 1820.
- ❖ **Missing Values:** I found that there are 733 null rows in the Code column. The null entries are regional aggregates such as "Low-income countries" and not country-specific. This may result in double counting.

Data cleaning operations:

- ❖ **Filtering Entities:** I removed all 733 classification entities (which have NULL codes) to concentrate only on specific country entities and the world entity for side-by-side comparisons.
- ❖ **Temporal Grouping:** Due to the sparse data distribution (1475-1820), I resorted to aggregate the data points into wider bins of decadal and centurial data to smooth out the timeline.
- ❖ **Deleting Defunct States:** I have explicitly removed the USSR from my dataset, since it's a political entity that no longer exist.
- ❖ **Consistency Check:** I carried out a verification step that ensured that for each row, the sum of the Illiteracy rate and Literacy rate was approximately 100%.

Methodology

- **Data Collection:** I manually downloaded the primary dataset from the Our World in Data (OWID) platform in .csv format.
- **Data Processing & Cleaning:** I have used **Python** for most of my tasks using **Pandas** library for the cleaning of the dataset, whereas **Seaborn** and **Matplotlib** libraries are used for creating the initial plots for analysis. And I also used **Datawrapper** for creating visualizations. [GitHub Repo Link](#)
- **Data Transformation:** By **aggregating**, I calculated the global mean values for literacy rate, By **grouping**, I grouped the data points into wider bins of decadal and centurial data to smooth out the timeline
- **Analytical Techniques:** I used **Trend Analysis** to track the change in a 12% literacy rate in 1820 to 87% today, and **Comparative Analysis** to find out which regions still lag below that global average.
- **Reproducibility:** The entire analysis can be reproduced using the Python code and the publicly available dataset.

Insights from the Data

The Inversion of Literacy (1475 vs. 2023): Global literacy has completely reversed over the past 500 years:

- ❖ The aggregated mean literacy rate was just 6.60% in 1475.
- ❖ The global average was 87.36% by 2023.

Mid-1960s: The Global Tipping Point: The World entity, which estimates the entire world population, crossed the 50% threshold in the mid-1960s, whereas the average of reporting countries did so in the 1910s due to a small sample of wealthy nations.

The 1950 "Zigzag": Between 1940 and 1950, the data reveals a sharp "dip" in the unweighted mean literacy:

- ❖ 1940: Based on just **seven** reporting nations, the average literacy rate seemed to be 60.2%.
- ❖ 1950: The sample size skyrocketed to **176** countries, but the mean literacy "dropped" to 52.3%.

Conclusion: The inclusion of a large number of developing countries in the global dataset for the first time created a statistical artifact rather than a real-world decline in education.

Modern Regional Laggards Significant geographic differences persist in the 21st century despite the global upward trend: A number of countries continue to significantly lag behind the 2023 global average of 87.36%.

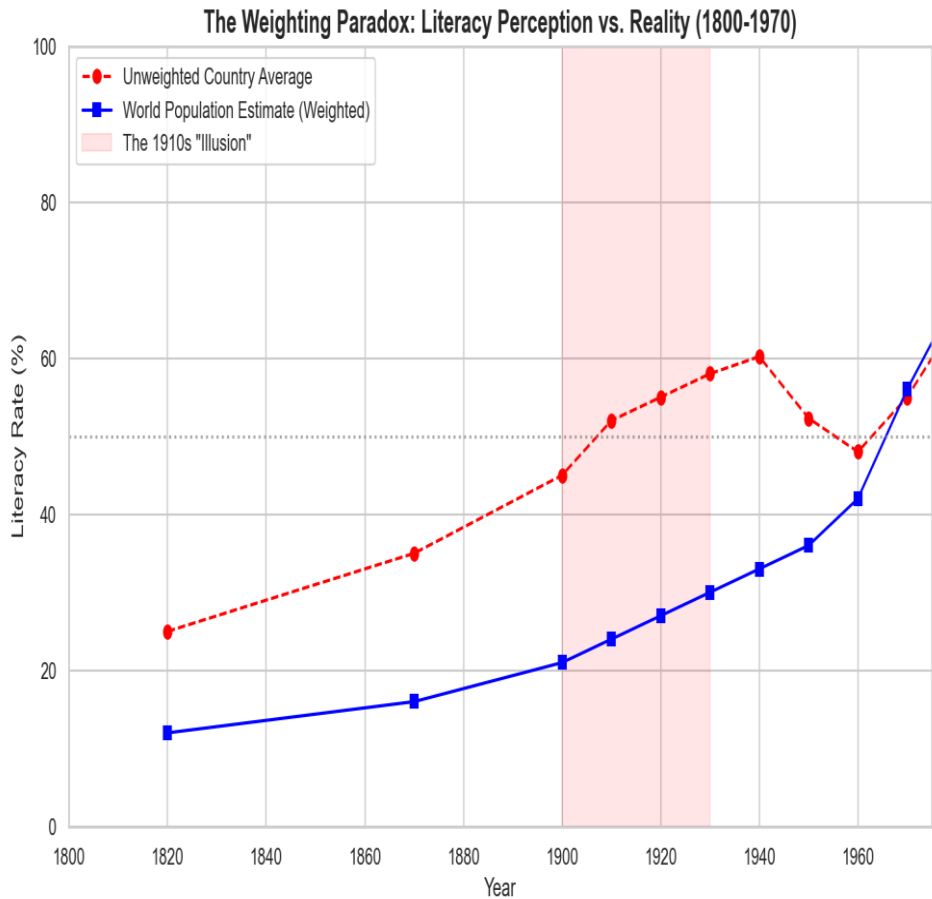
lowest rates observed (2018–2022): Chad: 30.6%, Mali: 35.0%, Niger: 35.6%, Afghanistan: 37.0%

The Great Acceleration (Post-1820): Stagnant Growth (1475–1820): The global literacy rate increased by just 5.45 percentage points over a period of 345 years. Exponential Growth (1950–2023): The global rate increased by 51.40 percentage points in just the last 73 years.

The "Statistical Illusion" of the 1910s and The True Global Milestone (The 1960s)

Visualization Description: A dual-line chart comparing the unweighted arithmetic mean of available countries against the population-weighted "World" estimate from 1800 to 1970.

- **Red Dashed Line:** Represents the unweighted average of reporting countries. It crosses the 50% threshold prematurely in the 1910s because the data was biased toward highly developed nations.
- **Blue Solid Line:** Represents the true global estimate (weighted by population). It shows a much slower, steadier climb, remaining below 50% until the mid-1960s.
- **The "Illusion" Zone:** Highlighted in light red, this area shows the five-decade gap where unweighted data incorrectly suggested a majority-literate world.



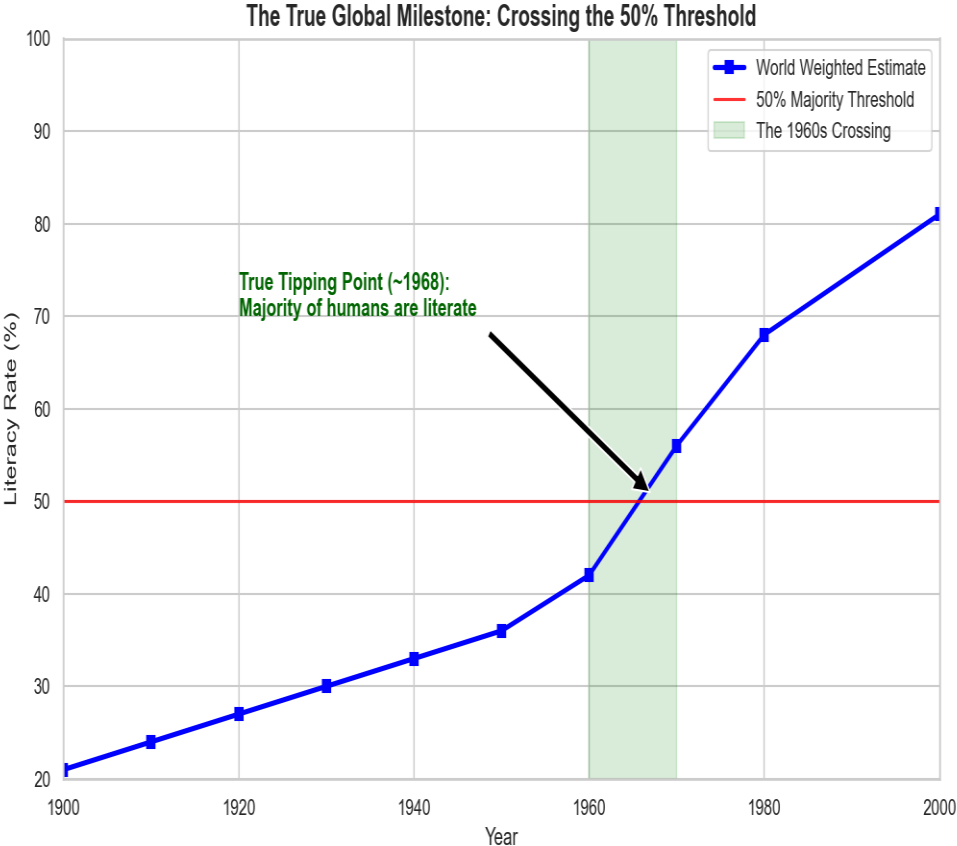
This chart is created with
Python(Matplotlib/Seaborn)

The "Statistical Illusion" of the 1910s and The True Global Milestone (The 1960s)

Visualization Description: A focused timeline chart (1900–2000) highlighting the precise moment the weighted global population estimate crossed the 50% majority threshold.

Tipping Point: The chart marks the mid-to-late 1960s (specifically ~1968) as the definitive moment in history when the majority of adults on Earth became literate.

Significance: The visual emphasizes the "Weighting Paradox," proving that once billions of people in the Global South were properly accounted for, the timeline for global development shifted significantly.



This chart is created with
Python(Matplotlib/Seaborn)

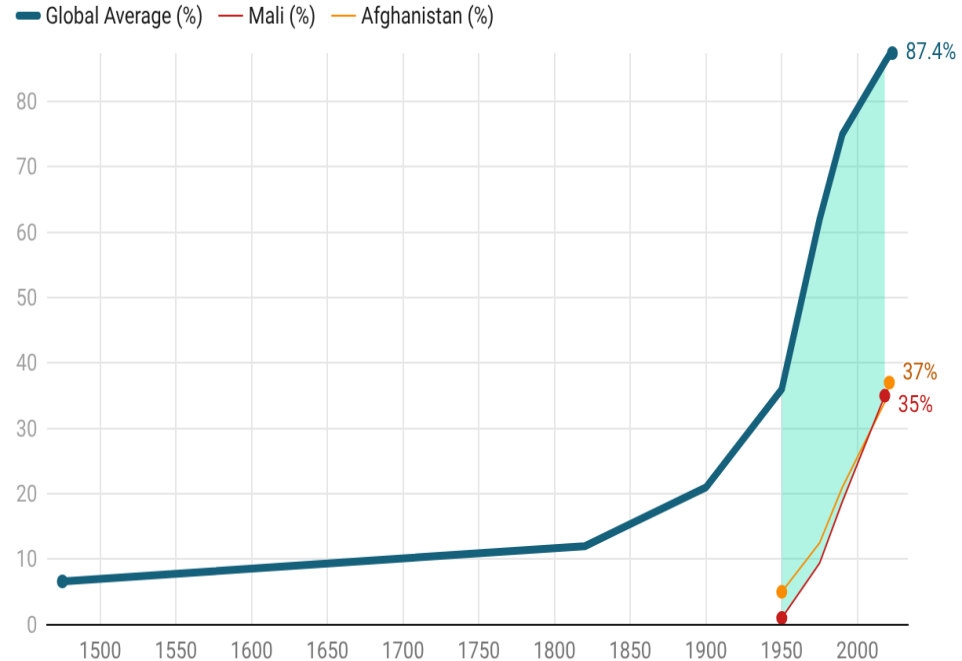
The Great Inversion vs. Regional Stagnation and Identifying the 21st-Century Laggards

Visualization Description: A "Velocity Chart" comparing the explosive growth of the global average against the much slower progress of "Laggard" nations.

- **The Global Surge:** Starting from 6.6% in 1475, the world reached a 50% "Tipping Point" in the late 1960s, eventually climbing to 87.4% today.
- **The Velocity Gap:** While the global rate accelerated 70 times faster post-1950 compared to the pre-industrial era, nations like Mali demonstrate a "low-literacy trap," only reaching 35.0% literacy by 2018, nearly 50 points behind the global mean, And Afghanistan reached 37.0% literacy by 2021.

The Great Inversion vs. Regional Stagnation

The Velocity Gap in Global Literacy (1475–2023)



Created with Datawrapper

The Great Inversion vs. Regional Stagnation and Identifying the 21st-Century Laggards

Visualization Description: A bar chart illustrating the "Final Mile" challenge by comparing the 2023 global average to the latest data from the world's most educational-divergent regions.

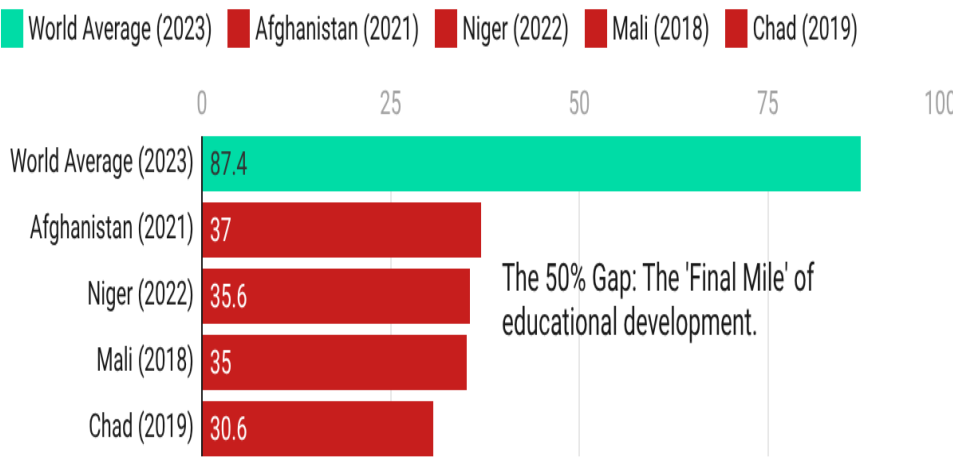
The Educational Divide: The gap between the global average and the lowest-performing nations is now approximately 50 to 57 percentage points.

Specific Laggards:

- World Average (2023): 87.4%
- Afghanistan (2021): 37.0%
- Mali (2018): 35.0%
- Chad (2019): 30.6%

The "Final Mile": Regional Laggards vs. Global Mean (2023)

While the global majority has crossed the 87% threshold, a critical group of nations remains over 50 percentage points behind.



Created with Datawrapper

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